HARD · ALGEBRA

Algebra

12

10 questions · ~15 min

SAT QUESTION BANK

2026-05-29

Q1 ALGEBRA

$$\frac{7}{2}x + 6y = 25$$

$$\frac{5}{2}x + 6y = 23$$

The solution to the given system of equations is x, y. What is the value of $\frac{17}{2}x + 18y$?

A. 2

B. 3

C. 48

D. 71

Q2

GRID-IN ALGEBRA

Line p is defined by $4y + 8x = 6$. Line r is perpendicular to line p in the xy-plane. What is the slope of line r?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q3

GRID-IN ALGEBRA

The functions f and g are defined as $f\left(x\right) = \frac{1}{4}x - 9$ and $g\left(x\right) = \frac{3}{4}x + 21$. If the function h is defined as $h(x) = f(x) + g(x)$, what is the x-coordinate of the x-intercept of the graph of $y = hx$ in the xy-plane?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q4 ALGEBRA

$$-12x + 14y = 36$$

$$-6x + 7y = -18$$

How many solutions does the given system of equations have?

A. Exactly one

B. Exactly two

C. Infinitely many

D. Zero

Q5 ALGEBRA

Keenan made 32 cups of vegetable broth. Keenan then filled x small jars and y large jars with all the vegetable broth he made. The equation $3x + 5y = 32$ represents this situation. Which is the best interpretation of $5y$ in this context?

A. The number of large jars Keenan filled

B. The number of small jars Keenan filled

C. The total number of cups of vegetable broth in the large jars

D. The total number of cups of vegetable broth in the small jars

Q6 ALGEBRA

$$Fx = \frac{9}{5}x - 273.15 + 32$$

The function F gives the temperature, in degrees Fahrenheit, that corresponds to a temperature of x kelvins. If a temperature increased by 2.10 kelvins, by how much did the temperature increase, in degrees Fahrenheit?

A. 3.78

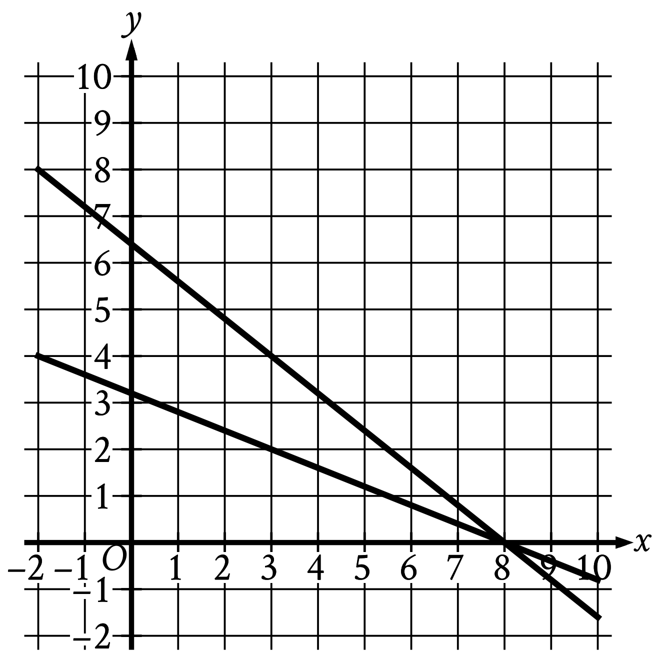
B. 35.78

C. 487.89

D. 519.89

Q7

ALGEBRA



- For the first line in the system:
    - The line slants gradually down from left to right.
    - The line passes through the following points:
        - (3 comma 2)
        - (8 comma 0)
- For the second line in the system:
    - The line slants gradually down from left to right.
    - The line passes through the following points:
        - (3 comma 4)
        - (8 comma 0)

What system of linear equations is represented by the lines shown?

A. $8x + 4y = 32$
$-10x - 4y = -64$

B. $8x - 4y = 32$

$-10x + 4y = -64$

C. $4x - 10y = 32$
$-8x + 10y = -64$

D. $4x + 10y = 32$
$-8x - 10y = -64$

$$5x + 7y = 1$$

$$ax + by = 1$$

In the given pair of equations, a and b are constants. The graph of this pair of equations in the xy-plane is a pair of perpendicular lines. Which of the following pairs of equations also represents a pair of perpendicular lines?

A. $10x + 7y = 1$
$ax - 2by = 1$

B. $10x + 7y = 1$
$ax + 2by = 1$

C. $10x + 7y = 1$
$2ax + by = 1$

D. $5x - 7y = 1$
$ax + by = 1$

Q9

ALGEBRA

$$7x - 5y = 4$$
$$4x - 8y = 9$$

If (x, y) is the solution to the system of equations above, what is the value of $3x + 3y$?

A. -13

B. -5

C. 5

D. 13

Q10 GRID-IN ALGEBRA

x	y
3	7
k	11
12	n

The table above shows the coordinates of three points on a line in the xy-plane, where k and n are constants. If the slope of the line is 2, what is the value of $k+n$?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.